

Particulate Matter Monitoring & Health Effects

Issue 2026-08-26 | Entry window 26 Aug 2026 | 14 in scope + 6 backfill, 36 out

14IN-WINDOW RECORDS
(36 SCREENED OUT)**7**CLUSTERS, EXACTLY
2 RECORDS EACH**9**INTERVAL-BEARING
ESTIMATES**6**OUT-OF-WINDOW
CALIBRATION BACKFILL**2**ABSTRACT-FREE
DEPOSITS (14%)

Signal of the day

The reference method that every automated PM monitor is traceable to is itself humidity-sensitive — and the EU's post-2030 uncertainty budget does not leave room for that.

An ISPRA-organised national interlaboratory comparison at Castel Romano put the EN 12341 gravimetric reference method for PM₁₀ and PM_{2.5} in front of **18 Italian regional environmental agencies** simultaneously. Systematic deviations between laboratories were driven **primarily by relative humidity and precipitation**, which compromise PM thermodynamic and mass stability during handling and conditioning, with a smaller filter-matrix contribution. Gravimetry is the sole traceability source for every automated measurement system, so this bias propagates downward into the entire monitoring chain — and **Directive (EU) 2024/2881 sharply tightens AMS residual uncertainty budgets for daily and annual PM₁₀ and PM_{2.5} from 2030**, at concentrations that are themselves declining. The uncomfortable implication for low-cost sensor work: the humidity artefact that every LCS calibration paper treats as a sensor problem is partly present in the reference the sensor is being calibrated *against*. Co-location residuals attributed to sensor RH response may contain a reference-side component that no amount of sensor-side modelling can remove.

What changed today

- ▶ **The cleanest component-resolved result this watch has carried.** Mao et al. ran 60,789 consecutive gastroscopies against gridded daily PM_{2.5} and its five chemical components. Total PM_{2.5} gave **IRR 1.076 (95% CI 1.024–1.132)** per IQR for endoscopically confirmed erosive gastric lesions, but the signal sits entirely in the **secondary inorganic fraction — ammonium 1.082 (1.026–1.141) and sulfate 1.070 (1.024–1.118)** — while nitrate, organic matter and black carbon are all flat at 1.012 with intervals spanning the null. Six estimates, one study, one exposure model, one outcome ascertainment: the internal comparison is as clean as this literature gets, and it says mass is the wrong metric.
- ▶ **Ultrafine particles get a mediation pathway.** Wang et al. modelled UFP at high spatiotemporal resolution for 7,569 Shanghai pregnancies and found per-IQR relative risks of 1.15–1.23 for hypertensive disorders of pregnancy and 1.29–1.41 for pre-eclampsia. First-trimester homocysteine mediated 2.51–8.87% of the UFP–pre-eclampsia association (**Hcy–PE RR 1.49, 1.04–2.13**), and the association held only in the low-vitamin-B₁₂ stratum. Mediated fractions under 10% are easy to over-read; the B₁₂ modification is the more actionable finding.
- ▶ **A large-cohort PM/BPH result with a stated threshold.** Wei et al. report 2.18% higher BPH risk per 1 µg/m³ PM_{2.5} across 123,037 UK Biobank men, replicated in a western-China cohort at 2.77%, with a restricted-cubic-spline **risk threshold at 9.83 µg/m³** — essentially the WHO interim target and the same ≥9 µg/m³ breakpoint the AF literature landed on last month. Thirty-four lipid metabolites are proposed as mediators, which at this cohort size is exploratory.
- ▶ **A dietary effect modification worth scepticism.** Kessler et al. report that a first-trimester IQR increase in PM was associated with a 0.08-unit decrease in birth-weight z-score at mean α-linolenic-acid intake but a **1.12-unit increase (0.19–2.05) at twice mean ALA intake** ($p_{\text{int}} = 0.04$). A sign reversal of that magnitude on $n = 297$, driven by the upper tail of a food-frequency-questionnaire distribution, is far more likely to be sparse-data bias than a real protective interaction. Reported here because it will be cited.
- ▶ **Six-record coverage backfill on low-cost sensor calibration** (see the sidebar). A Consensus topical sweep surfaced six 2026 calibration papers deposited between January and June — before the Crossref by-ISSN leg existed — that no leg of this pipeline has ever seen. They are *not* counted in the metric strip, the figures or the register, which describe the 26 August window only.
- ▶ **Two abstract-free deposits, both squarely on topic.** Condensable particulate matter in subzero remote Canadian mountain communities (*Environ Int*) and source-resolved PM_{2.5} oxidative potential in a shipbuilding complex (*Atmos Pollut Res*) arrived with title and DOI only. The Elsevier abstract backlog now spans four weeks and both go on the retry list at the top.

Corpus at a glance

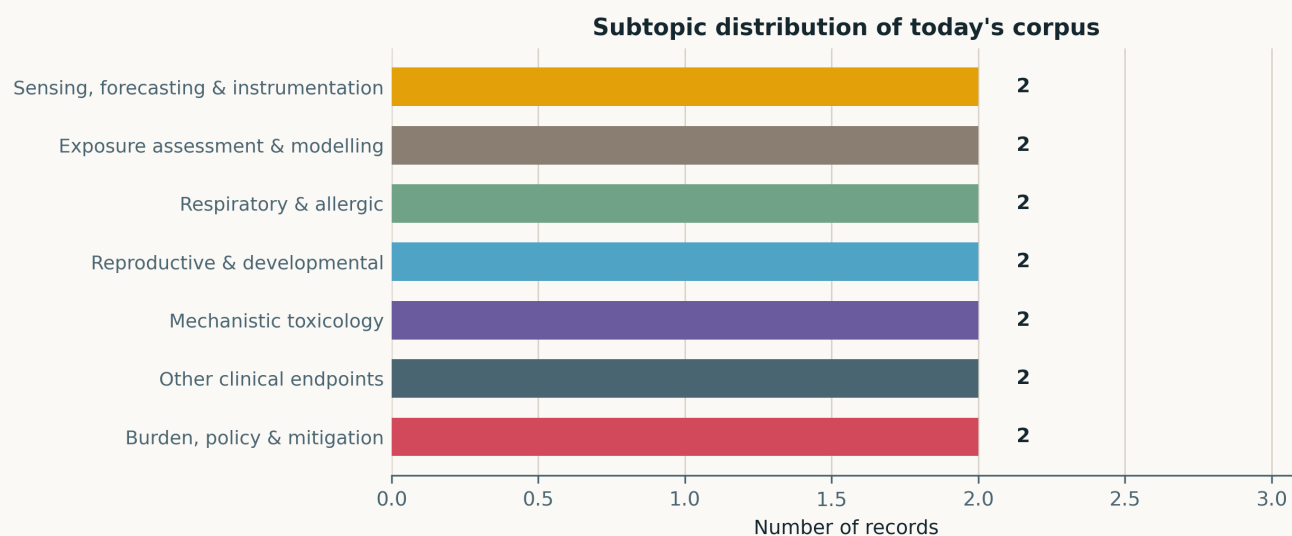


Fig. Records per subtopic cluster. **Seven clusters, exactly two records each** — a flat distribution this watch has not produced before, and an artefact of a single *Environment & Health* issue release landing four unrelated endpoint papers in one day rather than of any convergence in the field. *Cardiovascular & metabolic*, *Neuro / mental health* and *Occupational & indoor* are empty.

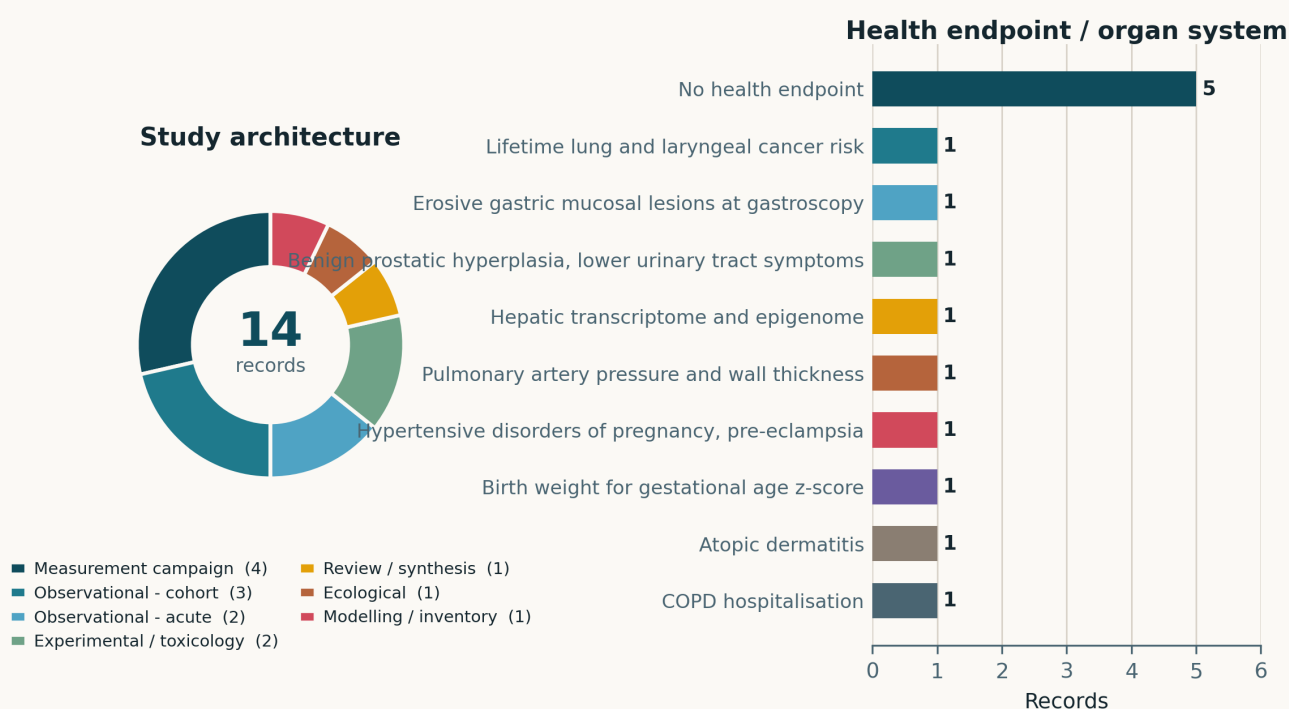


Fig. Left — study architecture as binned by the coarse grouper. *Measurement campaign* 4, *Observational — cohort* 3, *Observational — acute* 2, *Experimental / toxicology* 2, and one each for *Review / synthesis*, *Ecological* and *Modelling / inventory*. The measurement-campaign bar is heterogeneous: it holds an interlaboratory gravimetric comparison, an electron-microscopy characterisation and two abstract-free field records whose design is inferred from the title. Right — health endpoint. **Five records carry no human health endpoint** (two instrument, two exposure, one policy); the remaining nine endpoints are all singletons.

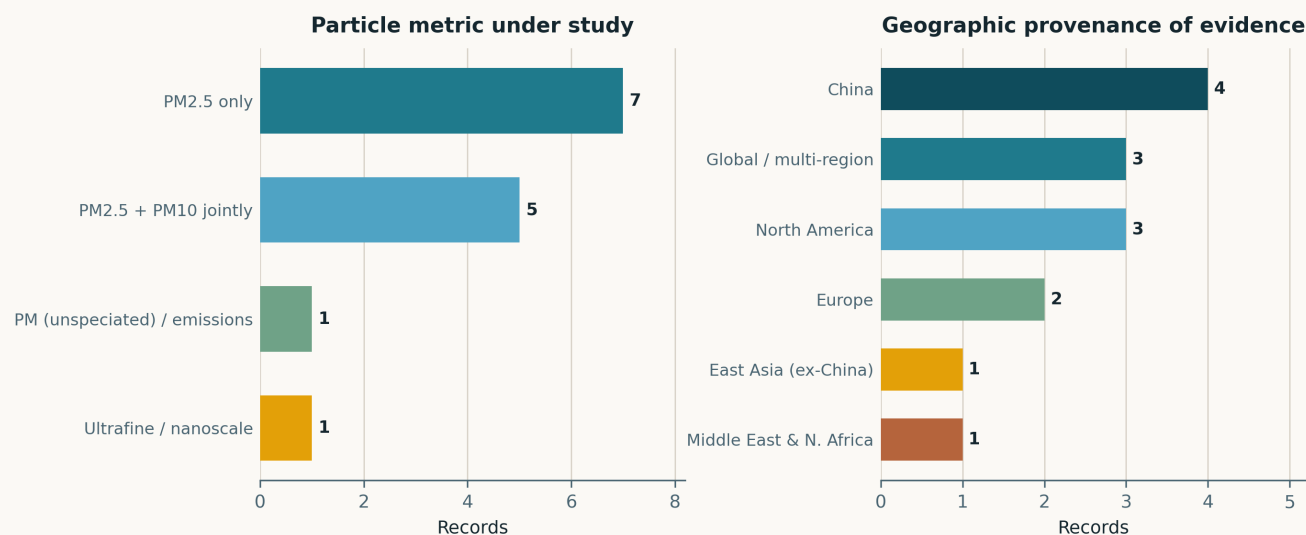


Fig. Left — particle metric. $PM_{2.5}$ only 7 and $PM_{2.5} + PM_{10}$ jointly 5; one record measures ultrafine particle number (Wang et al., the Shanghai pregnancy cohort) and one is unspiciated. Note that Mao et al. is binned as $PM_{2.5}$ -only although its actual exposure is five resolved chemical components — the grouper has no speciation bin, which is a real limitation given that speciation is where this window's signal lives. Right — geographic provenance: China 4, North America 3, multi-region syntheses 3, Europe 2, and one each from the Middle East & North Africa (Saudi Arabia) and East Asia outside China (South Korea). Nothing from Africa south of the Sahara, South Asia or Latin America.

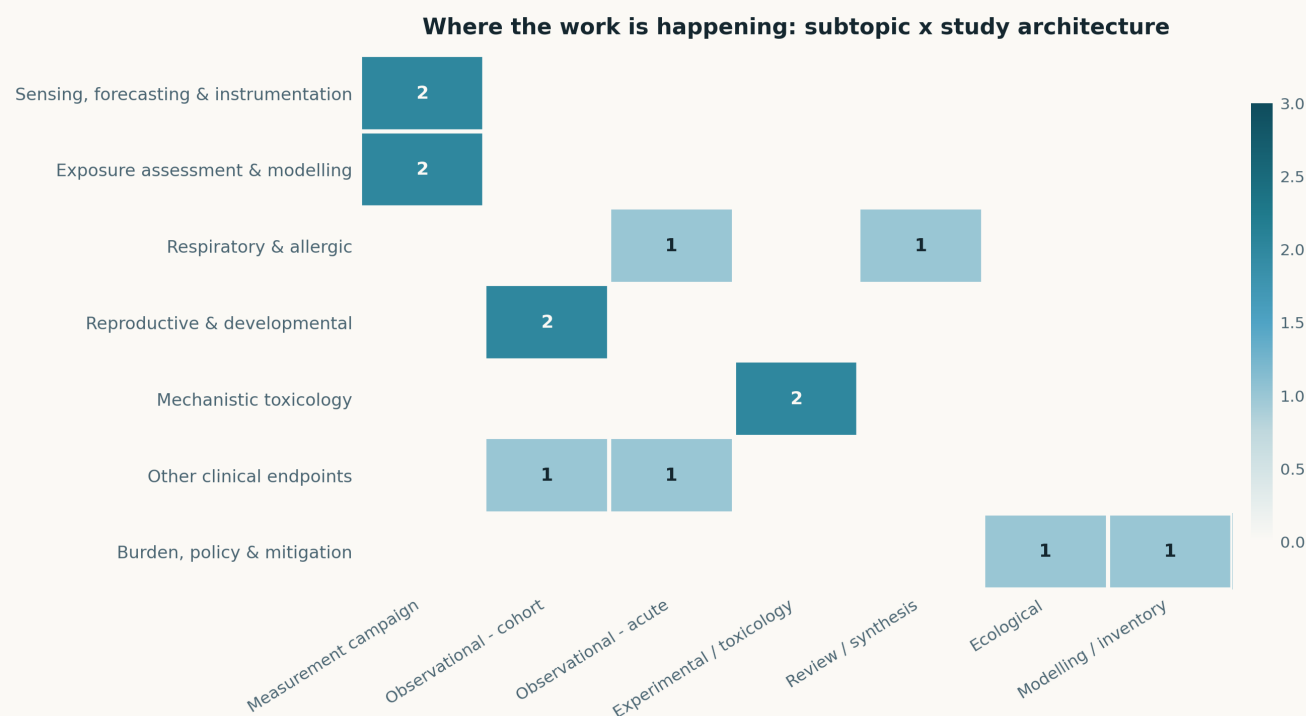


Fig. Subtopic × study-architecture cross-tabulation. With 14 records spread over seven clusters and seven architectures the matrix is almost entirely singletons. Four cells hold 2 — Sensing and Exposure assessment both × Measurement campaign, Reproductive & developmental × Observational — cohort, and Mechanistic toxicology × Experimental / toxicology — and every other occupied cell holds 1. The two architecture columns that carry no health endpoint anywhere (Measurement campaign, Modelling / inventory) account for 5 of the 14 records. Read as a sparsity map rather than as structure.

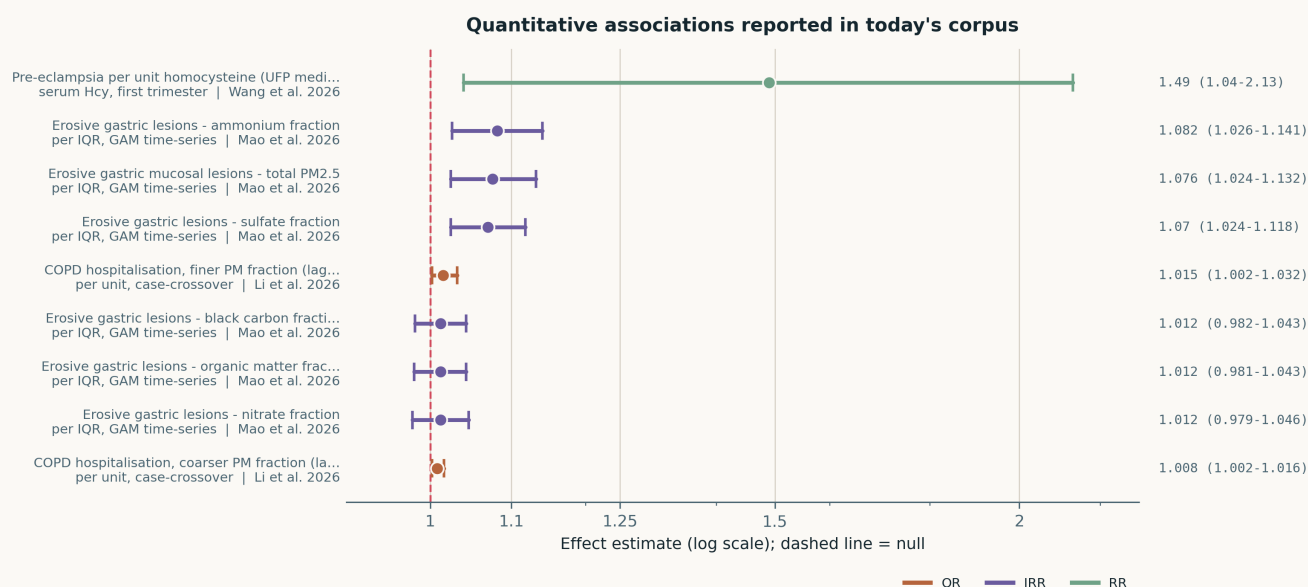


Fig. Nine interval-bearing estimates from three studies. **The six Mao et al. estimates are mutually comparable** — same cohort, same exposure model, same outcome, same per-IQR contrast — and are the only set in this watch's history for which a component-versus-component reading is legitimate. The two COPD odds ratios are on a per-unit rather than per-IQR contrast and are not comparable to them; note also that the publisher's indexing stripped the subscripts from the source abstract, so the assignment of finer and coarser PM fraction follows the order in which the abstract lists them and should be confirmed against the full text. The homocysteine relative risk is a *mediator-outcome* estimate, not an exposure-outcome estimate, and belongs on this axis only in the weak sense that it shares a ratio scale.

Life-course windows addressed today (records may span several stages)

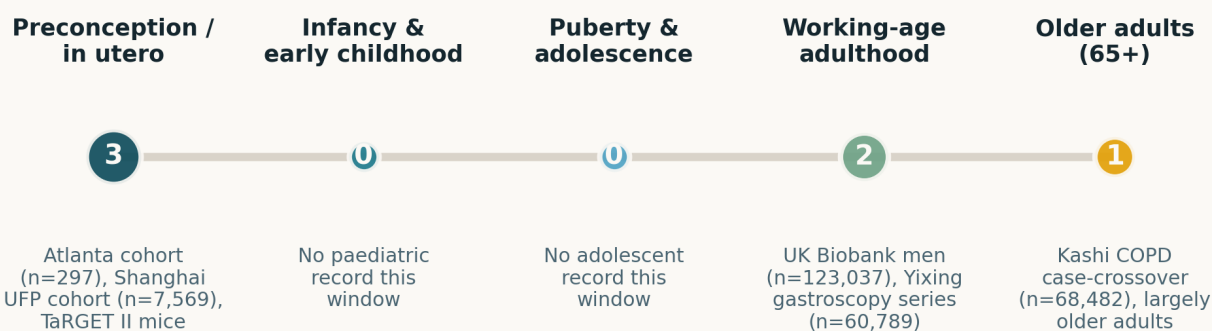


Fig. Life-course windows addressed, counted only where a record's study population is explicitly anchored to a stage. Six records qualify. **The preconception/in-utero window is occupied for the first time since 22 August** (3 records, after three consecutive empty issues), and the 65+ window carries the Kashi COPD series, whose population is largely older adults. Paediatric and adolescent remain empty — the paediatric window has now been empty for four consecutive issues.

Coverage backfill: low-cost sensor calibration

Six 2026 papers this pipeline never saw — out of window, not counted anywhere else in this issue

A Consensus topical sweep on low-cost sensor calibration, drift, co-location and humidity correction returned six 2026 records, **all six absent from seen.json**. Crossref created dates run 23 January to 21 June 2026 — that is, before the by-ISSN sensing leg was built on 22 August, and in journals (*Meas Sci Technol*, *Aerosol Sci Technol*) that leg still does not track. This is a genuine historical coverage hole, not a duplicate. They are recorded in seen.json so they will not re-ship, and both untracked ISSNs have been added to the harvester.

- ▶ **Ginsburg et al., *Aerosol Sci Technol* — 1,013 PurpleAir sensors** co-located within 500 m of reference monitors on six continents, January 2023 to March 2025. Raw PurpleAir overestimates $PM_{2.5}$ in most regions and fails EPA sensor performance targets; **US-derived corrections do not transfer outside North America**; a global quadratic helps in Australia/New Zealand and hurts elsewhere; **regional calibrations meet EPA 24 h thresholds in multiple regions and lift 24 h AQI category correctness to $\geq 77\%$** . The single most consequential record in this backfill for anyone running a non-US network. 10.1080/02786826.2026.2676293
- ▶ **Kaur et al., *Atmos Meas Tech* — 109 PMS5003/6003 sensors** at or near 75 US EPA sites, January 2017 to May 2025. Two thresholds built from relative humidity and the sensor's own coarse-to-submicron mass ratio detect dust-dominated periods *from sensor data alone*, using the trailing 336 hourly values that PurpleAir already archives. Applied as a non-linear correction to `pm2.5_alt`, this **cut mean bias error by $\sim 50\%$ on 97 sensors and RMSE by $\sim 30\%$ on 84**. Directly implementable with no additional hardware. 10.5194/amt-19-1077-2026
- ▶ **Gupta et al., *Aerosol Air Qual Res* — PA-II against FEM in Raleigh and Delhi, 2–4 week deployments** chosen to span contrasting aerosol loadings. Calibrated output reaches **within 12 % mean absolute bias hourly and 6 % daily**. The authors conclude pre-deployment regional calibration is a precondition for scientific use, which is the same conclusion Ginsburg et al. reach from a global dataset. 10.1007/s44408-026-00146-x
- ▶ **Zhivkov et al., *Atmosphere* — the calibration-transfer problem stated properly:** what happens to a machine-learning correction applied to a sensor that is nowhere near a reference station. 24 months, Sofia, 5 reference stations and 22 sensors of which 4 are co-located. Random Forest reached $R^2 \in (0.53, 0.75)$, and leave-one-station-out cross-validation yielded **uncertainty growth of 3.84–5.62 % per km** from the nearest reference (significant for PM_{10} and O_3). **Co-location sites 6 km apart hold midpoint uncertainty growth under 30 %**, which is an actionable network-siting rule. 10.3390/atmos17040335
- ▶ **Qian et al., *Atmos Meas Tech* — AutoML calibration for indoor $PM_{2.5}$** , staged through an intermediate drift-correction reference and fitted separately for clean-air and pollution-event concentration ranges. Chamber evaluation on two sensor models gave $R^2 > 0.90$ **with NRMSE and sMAPE roughly halved**. The split-range design is the transferable part; chamber validation limits the field claim. 10.5194/amt-19-603-2026
- ▶ **Feng et al., *Meas Sci Technol* — review of machine-learning, physics-based and hybrid physics-informed calibration.** Conclusion worth quoting: **no single strategy is universally optimal** — ML wins on predictive performance, physics on interpretability and physical consistency, hybrids on transferability — and non-linear models reaching sub- $10 \mu g m^{-3}$ error still face embedded-deployment, training-data-representativeness and region-specificity barriers. 10.1088/1361-6501/ae6298

Paper-level digest

Ordered by cluster; the 26 August window only. The six backfill records are covered in the preceding section and are not repeated here.

Sensing, forecasting & instrumentation

2 records

Anon et al. 2026 Atmosphere

Metrological consistency of the EN 12341 gravimetric reference method

National interlaboratory comparison run by ISPRA at Castel Romano with **18 Italian regional environmental agencies** sampling side by side, motivated by the absence of Certified Reference Materials for PM and by known sampler-design discrepancies. Systematic between-laboratory deviations were driven mainly by **relative humidity and precipitation** acting on PM thermodynamic and mass stability, with a minor filter-matrix effect. Because gravimetry is the only traceability source for automated measurement systems, the bias propagates into every AMS calibrated against it — and Directive (EU) 2024/2881 tightens AMS residual uncertainty budgets sharply from 2030, at concentrations that are falling. No numeric bias magnitudes appear in the abstract, which is the main gap; the full text is needed to size the effect. Crossref lists no first author.

doi: [10.3390/atmos17090828](https://doi.org/10.3390/atmos17090828)

Alharbi et al. 2026 Atmosphere

Morphology, surface chemistry and mineralogy of coastal PM_{2.5}, Jeddah

Three coastal sites over summer 2017 characterised by SEM, EDS mapping, XPS and XRD, linking bulk concentration (**22.2, 18.9 and 14.2 µg/m³** north to south) to particle morphology, surface oxidation state and crystalline phase. The authors are unusually candid about their own artefact: **borosilicate glass-fibre substrates dominate the SEM field**, so aerosol is resolved only as discrete sub-micrometre particles and agglomerates decorating the fibres, and the morphological interpretation is framed accordingly. That admission is worth more than the result — filter-substrate choice constrains what electron microscopy of a PM sample can establish, and most papers in this genre do not say so.

doi: [10.3390/atmos17090830](https://doi.org/10.3390/atmos17090830)

Exposure assessment & modelling

2 records

Anon et al. 2026b Environ Int

Condensable particulate matter and severe winter PM_{2.5} in remote Canadian mountain communities

Metadata-only deposit — title and DOI without an abstract. The topic is directly relevant and under-covered: condensable PM forms downstream of the stack as vapour cools, is invisible to filter-based source testing conducted at temperature, and in subzero valley-inversion conditions can dominate ambient PM_{2.5} in communities with no industrial source of any size. If the finding is what the title implies, it also bears on optical sensing — semi-volatile condensable material has different scattering and hygroscopic behaviour from the combustion soot most correction factors assume. Carried at tier C and flagged as the top abstract retry.

doi: [10.1016/j.envint.2026.110472](https://doi.org/10.1016/j.envint.2026.110472)

Anon et al. 2026c Atmos Pollut Res

Source-specific PM_{2.5} oxidative potential in a national shipbuilding complex

Metadata-only deposit. The title asserts that metal processing and combustion dominate source-apportioned oxidative potential, which if supported is another instance of the recurring result that OP-weighted and mass-weighted source contributions rank differently. No apportionment method, OP assay, site count or period is available. Tier C, second on the abstract retry list.

doi: [10.1016/j.apr.2026.103189](https://doi.org/10.1016/j.apr.2026.103189)

Reproductive & developmental

2 records

Wang et al. 2026 Environ Health (Wash)*Ultrafine particles, hypertensive disorders of pregnancy, and homocysteine mediation*

Shanghai birth cohort, **7,569 pregnancies**, 2017–2018, with individual UFP exposure from a high-spatiotemporal-resolution prediction model at residential address and serum homocysteine, folate and B₁₂ measured in early pregnancy. Poisson regression with robust error variance gave per-IQR RR **1.15–1.23 for hypertensive disorders of pregnancy and 1.29–1.41 for pre-eclampsia** across the prepregnancy, first- and second-trimester windows. Homocysteine was associated with pre-eclampsia (RR 1.49, 1.04–2.13) and mediated 8.87% and 2.51% of the prepregnancy and first-trimester associations. Stratification showed the UFP-PE association only in the low-B₁₂ group. Two cautions: the exposure is modelled, not measured, so UFP-PM_{2.5} collinearity is not resolvable here; and mediated fractions this small are within the range that measurement error in the mediator can manufacture.

doi: 10.1021/envhealth.5c00760

Kessler et al. 2026 Environ Health (Wash)*Prenatal traffic-related air pollution, dietary PUFA intake and birth weight*

Atlanta Maternal–Child Cohort, **297 pregnant African American women**, first-trimester traffic-related CO, PM_{2.5} and NO_x crossed with dietary polyunsaturated fatty acid intake in a joint mixture model. At mean α -linolenic acid intake, a per-IQR PM increase was associated with a **0.08-unit decrease (0.02–0.18) in birth-weight-for-gestational-age z-score**; at twice mean ALA intake the same increase was associated with a **1.12-unit increase (0.19–2.05)**, $p_{\text{int}} = 0.04$, with an almost identical pattern for linoleic acid. The authors read this as PUFA mitigation. A cleaner reading is that a 14-fold sign-reversing interaction estimated on 297 women in the sparse upper tail of a food-frequency distribution is the signature of sparse-data bias. Worth tracking because the mechanistic story is plausible and the result will be cited widely.

doi: 10.1021/envhealth.5c00490

Other clinical endpoints

2 records

Mao et al. 2026 Front Public Health*Component-specific PM_{2.5} and endoscopically confirmed erosive gastric lesions*

Consecutive gastroscopies at Yixing People's Hospital, 1 Jan 2024 to 31 Dec 2025: **60,789 individuals, 14,325 with erosive gastric mucosal lesions**, against gridded daily PM_{2.5} and five chemical components in generalized additive models with attributable-burden analysis. Per IQR, **total PM_{2.5} IRR 1.076 (1.024–1.132)**; after FDR correction only **ammonium 1.082 (1.026–1.141)** and **sulfate 1.070 (1.024–1.118)** survived, with **nitrate, organic matter and black carbon all at 1.012** and intervals crossing the null. Endoscopic ascertainment removes the symptom-reporting bias that weakens most digestive air-pollution studies. The limitation is real and stated: the five components are strongly collinear and share one exposure model, so their independent contributions are not identified. Still the sharpest secondary-inorganic-versus-carbonaceous contrast available.

doi: 10.3389/fpubh.2026.1891236

Wei et al. 2026 Environ Health (Wash)*Air pollution, lipid mediators and benign prostatic hyperplasia*

UK Biobank, **123,037 men**, with a western-China cohort replication. **2.18% higher BPH risk per 1 $\mu\text{g}/\text{m}^3$ PM_{2.5}** ($p = 0.0029$) in UK Biobank and 2.77% in the replication; mixture exposure by WQS and qgcomp gave 3.53% and 2.20% per unit for BPH and 6.78% and 9.17% for lower urinary tract symptoms. Restricted cubic splines put the **BPH risk threshold at 9.83 $\mu\text{g}/\text{m}^3$** , and joint high-PM/high-fat-diet exposure carried 16% higher risk than low/low. Thirty-four lipid metabolites are nominated as mediators. Two problems: BPH ascertainment in UK Biobank is by linked diagnosis and is sensitive to healthcare-contact patterns that correlate with urbanicity and therefore with PM_{2.5}; and 34 mediators from an untargeted panel without a stated multiplicity correction is hypothesis generation, not mediation analysis.

doi: 10.1021/envhealth.5c00806

Respiratory & allergic

2 records

Li et al. 2026 Int J Chron Obstruct Pulmon Dis*Air pollution and COPD hospitalisation in Kashi, Xinjiang*

Time-stratified case-crossover on **68,482 COPD patients** in Kashi, 2012–2022, with daily pollutant concentrations matched to residential address rather than to a single city monitor. Strongest PM association at **lag day 4**: odds ratios **1.015 (1.002–1.032)** and **1.008 (1.002–1.016)** for the two PM fractions, with NO₂ and CO both at 1.034. Young patients, those with prior severe exacerbations, farmers, smokers and cool-season admissions were the susceptible strata. The honest part is the sensitivity analysis: **excluding COVID-19-period hospitalisations left the PM and NO₂ associations barely significant and CO non-significant**, which means a material share of the headline estimate rides on pandemic-era admission patterns. The publisher's indexing stripped every subscript from the abstract, so which PM fraction is which must be read from listing order. Kashi sits in a severe dust-storm corridor, making this one of the few large COPD series in a mineral-dust-dominated aerosol.

doi: 10.2147/COPD.S617521

Wang & Li 2026 Front Public Health*Air pollution and atopic dermatitis*

Narrative review of PubMed, Web of Science and CNKI, April 2021 to December 2025, restricted to peer-reviewed work with impact factor ≥ 3.0 — a screening criterion that is a proxy for nothing and silently excludes most dermatology and most regional epidemiology. Covers epidemiology, skin-barrier and microbiome mechanisms, climatic and geographical disparity, and prevention. No quantitative synthesis and no effect estimates. Carried at tier C as a citation anchor for the cutaneous endpoint, which is otherwise absent from this watch.

doi: 10.3389/fpubh.2026.1854212

Mechanistic toxicology

2 records

Liu et al. 2026 Environ Health (Wash)*PM_{2.5}-induced pulmonary vascular remodelling via MSC-mediated endothelial senescence*

Real-ambient PM_{2.5} exposure in mice plus a pulmonary arterial endothelial cell model. Exposure raised pulmonary artery pressure and arterial wall thickness in vivo and induced senescence and mitochondrial dysfunction in mouse pulmonary arterial endothelial cells. Bioinformatic screening nominated the transcription factor MSC as the regulator; **MSC overexpression rescued mitochondrial-dysfunction-associated senescence**, and MSC was shown to act through the downstream target NDUFB3, a complex I subunit — which makes the mitochondrial phenotype mechanistically coherent rather than merely correlative. **Real-ambient rather than instilled exposure** keeps the dosimetry defensible. No concentration, duration or particle characterisation appears in the abstract, so comparability to other real-ambient systems cannot be judged.

doi: 10.1021/envhealth.5c00595

TaRGET II Consortium 2026 bioRxiv (preprint)*Lifecourse sex-specific molecular response to early-life toxicant exposure*

Consortium-scale design: mice exposed from two weeks pre-conception through gestation and lactation to arsenic, lead, tributyltin, BPA, DEHP, TCDD or PM_{2.5}, with **1,025 omics datasets** generated from liver across three life stages. The headline is a significant sex-biased transcriptomic and epigenomic response to a common exposure. For this watch the value is comparative — PM_{2.5} is run alongside six reference toxicants under one protocol, which is the only way to judge whether a PM molecular signature is distinctive or generic. Liver-only, and mouse inhalation dosimetry does not scale simply to humans. Unreviewed.

doi: 10.64898/2026.08.20.746014

Burden, policy & mitigation

2 records

Anon et al. 2026d Prev Med Rep*Lifetime risk of respiratory-system cancers across 185 countries*

GLOBOCAN 2022 combined with UN life tables under an adjusted-for-multiple-primaries framework. **Global lifetime risk of developing lung cancer 4.92 % and of dying from it 4.08 %**, against 0.33 % and 0.20 % for laryngeal cancer, with a steep gradient from **0.34 % in low-HDI to 5.45 % in very-high-HDI countries**. All four pollutants were positively associated with lifetime risk in univariable linear regression, PM_{2.5} strongest. Two hard limits: the HDI gradient in lung cancer incidence is dominated by historical smoking prevalence and by diagnostic capacity, and country-level pollution from **OpenWeather** is a consumer weather API rather than a validated exposure product. Univariable ecological regression across 185 countries cannot separate PM from GDP, smoking or registry completeness. Tier B for the burden estimates; the pollution association should not be cited as evidence of effect.

doi: 10.1016/j.pmedr.2026.103602

Anon et al. 2026e Res Sq (preprint)*A Harmonization Index for industrial PM emission regulations*

A Regulatory Coding Framework converting legally binding documents into standardized scores across **21 indicators in three dimensions** — regulatory stringency, institutional design and enforcement capacity — aggregated into a composite Harmonization Index, applied to the EU, US, China and Uzbekistan as exemplars of technology-based, source-category-based, hybrid and dispersion-based regulatory architectures. The problem is genuine: industrial PM limits are not comparable across jurisdictions because they are not the same kind of object. The method is documentary coding of what is written, which measures statutory text and not emissions; the enforcement-capacity dimension is the one that would need external validation and is the one hardest to score from documents. Unreviewed, four jurisdictions, tier C.

doi: 10.21203/rs.3.rs-10491363/v1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Reproductive & developmental					
Kessler et al.	Joint Mixture Associations of Prenatal Air Pollution and Polyunsaturated Fatty Acid Intake on Birth Weight Outcomes in an African American Cohort	<i>Environ Health (Wash)</i>	Prospective birth cohort (mixtures)	First-trimester traffic-related PM _{2.5} , CO, NOx	United States
Wang et al.	Ambient Ultrafine Particles and Hypertensive Disorders of Pregnancy: A Mediation Analysis of Homocysteine in a Birth Cohort	<i>Environ Health (Wash)</i>	Prospective birth cohort	Modelled ultrafine particle number, high spatiotemporal resolution	China
Respiratory & allergic					
Li et al.	Associations Between Air Pollution and Hospitalization of Patients with Chronic Obstructive Pulmonary Disease in Kashi, China: A Time-Stratified Case-Crossover Study	<i>Int J Chron Obstruct Pulmon Dis</i>	Time-stratified case-crossover	Address-matched daily PM _{2.5} and PM ₁₀	China
Wang & Li	Advances in research on the association between global air pollution and atopic dermatitis	<i>Front Public Health</i>	Narrative review	Ambient PM _{2.5} /PM ₁₀ and gaseous co-pollutants	Multi-region
Mechanistic toxicology					
Liu et al.	Ambient PM _{2.5} Induced Pulmonary Vascular Remodeling via MSC-Mediated Cellular Senescence in Pulmonary Arterial Endothelial Cells	<i>Environ Health (Wash)</i>	Real-ambient murine + cell model	Real-ambient PM _{2.5} exposure	China
TaRGET II Consortium	Lifecourse sex-specific molecular response to early-life exposures of toxic substances	<i>bioRxiv (preprint)</i>	Multi-omics animal consortium	PM _{2.5} among seven toxicants, preconception through lactation	United States
Sensing, forecasting & instrumentation					
Alharbi et al.	Multi-Technique Characterization of Atmospheric Aerosol Particles from the Coastal Area of Jeddah, Saudi Arabia: Morphology, Surface Chemistry, and Mineralogy	<i>Atmosphere</i>	Field campaign	PM _{2.5} morphology, surface oxidation state, crystalline phase (SEM/EDS/XPS/XRD)	Saudi Arabia
Anon et al.	Assessment Campaign of the Gravimetric Method (PM ₁₀ and PM _{2.5}): Analyzing Environmental Artifacts and Metrological Consistency for the 2030 Regulatory Challenge	<i>Atmosphere</i>	Interlaboratory comparison	Gravimetric PM ₁₀ and PM _{2.5} (EN 12341)	Italy
Exposure assessment & modelling					
Anon et al.	Severe winter PM _{2.5} pollution in remote Canadian mountain communities: Insights into the role of condensable particulate matter under subzero conditions	<i>Environ Int</i>	Field campaign	PM _{2.5} including condensable fraction, subzero conditions	Canada
Anon et al.	Metal Processing and Combustion Dominate Source-Specific PM _{2.5} Oxidative Potential in a National Shipbuilding Complex	<i>Atmos Pollut Res</i>	Field campaign	Source-apportioned PM _{2.5} oxidative potential	South Korea
Burden, policy & mitigation					
Anon et al.	Lifetime risk of respiratory system cancers across 185 countries: Global variation, socioeconomic inequalities, and ecological associations with air pollution	<i>Prev Med Rep</i>	Ecological (GLOBOCAN secondary analysis)	Country-level annual PM _{2.5} , PM ₁₀ , NO ₂ , SO ₂	Multi-region
Anon et al.	A Harmonization Index for Comparative Assessment of Industrial Particulate Matter Emission Regulations	<i>Res Sq (preprint)</i>	Regulatory coding framework	Industrial PM emission limits across regulatory architectures	Multi-region
Other clinical endpoints					
Mao et al.	Component-specific effects of PM _{2.5} on endoscopically identified erosive gastric mucosal lesions: a time-series study in Yixing, China	<i>Front Public Health</i>	Time-series (generalized additive model)	Gridded daily PM _{2.5} and five chemical components	China
Wei et al.	Prospective Metabolomic Cohort of 123,037 Men Reveals Lipid Mediators Linking Air Pollution to Benign Prostatic Hyperplasia	<i>Environ Health (Wash)</i>	Prospective cohort (mixtures)	Annual PM _{2.5} , PM ₁₀ , NO ₂ , NOx; WQS and qgcomp mixtures	United Kingdom

Provenance and method

Entry window **2026-08-26 to 2026-08-26**, a single day of deposits. **Sources.** PubMed E-utilities on [Date - Entry], run as two separate queries (health axis 11, instrumentation axis 0). Europe PMC CREATION_DATE 12. Crossref by-ISSN across the tracked non-PubMed-indexed titles, 33 deposits after front-matter suppression. OpenAlex remains broken and was not queried. arXiv contributed nothing new. **Consensus** was run on the low-cost-sensor calibration axis and returned six 2026 records, all out of window — reported separately in the *Coverage backfill* section and excluded from every count, figure and register row in this issue. **Screening.** 56 raw records, 52 unique after PMID/DOI/title collapse, 2 already in *seen.json*, **14 retained and 36 rejected** with reason logged. Rejections were dominated by non-particle atmospheric science (PBL parameterisation, cloud microphysics, ammonia, BVOC flux, ionospheric TEC), non-airborne chemical exposure (mycotoxins, organophosphates, phenanthrene metabolites) and **eleven Elsevier retraction notices deposited on a single day**. **Two abstract-free deposits** are carried at tier C on title and journal alone and are labelled as such in the digest. **Effect estimates.** Nine estimates from three studies. Only the six Mao et al. component IRRs are mutually comparable; the COPD odds ratios use a different contrast and the homocysteine estimate is a mediator–outcome association. See the figure caption. **Harvester change made this run.** *Meas Sci Technol* (0957-0233) and *Aerosol Sci Technol* (0278-6826) were added to the tracked-ISSN list, since both carried 2026 low-cost-sensor calibration work that no leg of this pipeline had seen. **Labels.** Subtopic, design, particle-metric and geography are single-label by dominant contribution — a judgement, not a controlled vocabulary. All DOI links were resolved before build.